

Assignment-3.2

“Diagnosing hardware failure”

TASK 1 List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

1. Defective Power Supply Unit (PSU) The power supply unit stops sending electrical power to the motherboard. Because no power reaches the system, the computer completely fails to turn on, showing no lights, no fan movement, and making no sound when you press the power button.

2. Faulty or Unseated RAM The RAM sticks have dirty pins, bad connections, or internal memory chips that fail to load data. The computer turns on and internal fans spin, but the monitor screen stays completely black without showing any startup text, accompanied by repeating warning beeps from the motherboard.

3. Overheating CPU The thermal paste between the processor and heat sink dries up, or the CPU cooling fan stops spinning due to dust buildup. The computer turns on and works normally for a few minutes, but then suddenly turns off by itself when you run heavy programs or games to protect the processor from melting.

4. Failing Hard Drive or SSD The internal magnetic platters inside a hard drive wear down over time, or the flash memory chips in an SSD reach their maximum read and write cycles. The computer becomes extremely slow, freezes during daily use, displays blue screen error messages, or gives a "Boot Device Not Found" error when powered on.

5. Failing Graphics Card (GPU) The video card processor overheats, or its dedicated video memory chips become corrupt. You see strange colored lines, flickering dots, and distorted blocks across your monitor screen, or the display loses signal completely whenever you launch a 3D game.

TASK 2 Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step 1: Test the Wall Power Socket Unplug the laptop adapter from the wall and plug in a working lamp or phone charger to ensure the wall socket is actually supplying electricity.

Step 2: Inspect the Charging Cable and Adapter Block Check the entire length of the charger cord for physical cracks, sharp bends, exposed wires, or burn marks, and feel if the adapter block warms up slightly when plugged in to ensure power is passing through it.

Step 3: Check the Laptop DC Charging Port Shine a light inside the charging port on the side of the laptop to look for dust buildup, trapped lint, or a loose center pin that prevents a solid physical connection with the charger tip.

Step 4: Perform a Hard Power Reset and Battery Test Remove the external battery if it is detachable, unplug the charger, hold down the power button for 30 seconds to drain leftover static electricity, and then plug in only the charger to see if the laptop turns on without the battery installed.

Step 5: Measure the Adapter Voltage Output with a Multimeter Set a multimeter to DC voltage, place the red probe inside the charger tip and the black probe against the outside metal ring, and check if the measured voltage matches the required voltage printed on the adapter label.

TASK 3 Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

1. Testing Procedure

Turn your multimeter dial to DC Voltage. Plug the USB phone charger into a live wall socket. Touch the red probe to the positive pin inside the USB port and the black probe to the ground pin.

2. Measured Voltage Reading

5.10V

3. Evaluation and Conclusion

The measured output of 5.10V falls within the standard safe USB voltage range of 4.75V to 5.25V. This confirms that the charger is working correctly and will safely supply power to a smartphone without causing damage.

TASK 4 A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

1. Turn off the power switch on the back of the power supply box. Unplug the main power cord from the wall outlet. Disconnect the large 24-pin power cable from the computer's motherboard.

2. Take a standard metal paperclip and bend it into a U-shape. Look at the 24-pin plastic connector and find the single **Green wire** (PS_ON). Push one end of the paperclip into the green wire slot, and push the other end into any **Black wire** (Ground) slot right next to it.

3. Plug the power cord back into the wall and flip the switch on the back of the power supply box to ON. The fan inside the power supply box should start spinning, showing that the unit is receiving power.

4. Turn the center dial of your multimeter to **DC Voltage** (V_{DC} or the 20V setting). Make sure the multimeter's black test lead is plugged into the COM port and the red lead is plugged into the V port.

5. Push the tip of the **Red multimeter probe** into any connector slot that has a **Yellow wire** (+ 12V power). Push the tip of the **Black multimeter probe** into any connector slot that has a **Black wire** (Ground). Read the number on the multimeter screen.

- **Hardware Failure Analysis** A voltage reading of 11.20V is below the minimum safe threshold of 11.50V required for a stable + 12V power rail. This low reading indicates that the internal capacitors or voltage regulators inside the power supply are degraded and failing. When the computer demands power for heavy tasks, the voltage drops even lower, causing the computer to shut down automatically to protect the motherboard, processor, and graphics card from electrical damage. The power supply unit must be replaced.